

Stage-3 LONO Ablation Regime Weighting After the Stage-2 μ -Anchor Winner

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Claim: after locking Stage 2 to the LONO winner `mu_anchor`, the Stage 3 regime/anchor ablation does *not* produce a large separation. The blended uncertain-regime variant gives the best mean RMSE/MAE, while `anchor_off` gives the best 2σ coverage.

This deck:

- fixes Stage 1 to `a_dp050_plus_pressures`;
- fixes Stage 2 to `mu_anchor` ($\lambda_\mu = 10^{-2}$, $\lambda_\sigma = 0$);
- compares four Stage 3 variants under the same five LONO folds;
- reports held-out metrics on each nozzle family's uncensored CDF P50 points;
- separates the small Stage 3 tuning effect from the much larger Nozzle0 OOD failure.

Validated run: `run_stage3_lono_ablation.py`, seed 42, 5 folds $\times$ 4 Stage 3 ablations.

Output: `MLP/runs_mlp/stage3_lono_ablation_20260521_092405/`

Experimental Contract

Fixed upstream choices

- Stage 1 feature family: `a_dp050_plus_pressures`
- Stage 2 teacher: `mu_anchor`, selected by the Stage 2 LONO ablation
- Seed: 42
- LONO folds: `Nozzle0`, `Nozzle2`, `Nozzle1`, `Nozzle5`, `Nozzle4`
- Evaluation: held-out nozzle's uncensored CDF P50 points

Stage-3 variant

Design change

baseline

baseline regime weighting with early anchor enabled

`raw_reliable_no_kd`

remove KD blending from reliable raw-CDF bins

`raw_uncertain_raw05_kd025`

blend raw supervision back into uncertain bins

`anchor_off`

remove the explicit Stage 3 early-time anchor

This is a downstream Stage 3 tuning test. It is not a cross-architecture SVGP comparison and it is not the later

LONO Results: Summary (Mean $\pm$ Std Across 5 Folds)

Ablation	RMSE [mm]	MAE [mm]	Bias [mm]	cov _{1σ}	cov _{2σ}
baseline	12.68 $\pm$ 12.74	9.97 $\pm$ 10.55	-7.86 $\pm$ 11.28	0.565	0.759
raw_reliable_no_kd	12.76 $\pm$ 12.48	10.01 $\pm$ 10.22	-7.45 $\pm$ 11.35	0.556	0.761
raw_uncertain_raw05_kd025	12.55 $\pm$ 12.46	9.80 $\pm$ 10.18	-6.87 $\pm$ 11.53	0.575	0.770
anchor_off	12.73 $\pm$ 12.44	9.86 $\pm$ 10.22	-7.11 $\pm$ 11.51	0.570	0.775

Reading the table

- Best mean RMSE/MAE: **raw_uncertain_raw05_kd025**.
- Best 2 σ coverage: **anchor_off**.
- The spread across Stage 3 variants is only 0.21 mm RMSE, much smaller than the cross-fold standard deviation.

Per-Fold RMSE: Where the Ranking Comes From

Ablation	Nozzle0 (311149)	Nozzle2 (60369)	Nozzle1 (25705)	Nozzle5 (33017)	Nozzle4 (28139)	Mean
baseline	35.31	8.57	7.20	4.61	7.72	12.68
raw_reliable_no_kd	34.99	8.24	7.37	5.20	7.99	12.76
raw_uncertain_raw05_kd025	34.76	7.74	6.61	5.52	8.13	12.55
anchor_off	34.92	7.76	7.16	5.60	8.20	12.73

- raw_uncertain_raw05_kd025 wins the two most informative modern-OOD folds outside Nozzle0: Nozzle2 and Nozzle1.
- baseline wins Nozzle5 and Nozzle4, so the Stage 3 effect is not monotonic.
- Nozzle0 remains a design-family extrapolation failure for every Stage 3 variant.

Nozzle0 Dominates the Variance

Nozzle0 carries **311,149** uncensored evaluation points and is the only first-generation nozzle family in the five-fold set. Every Stage 3 variant lands in the same failure band:

- RMSE 34.76–35.31 mm
- bias –26.78––27.62 mm
- 2σ coverage 0.140–0.171

Implication: Stage 3 regime weighting cannot repair a missing design family. The Nozzle0 fold is evidence about OOD geometry, not about the fine details of raw/KD mixing.

Fold	Points	Best RMSE
Nozzle0	311,149	34.76
Nozzle2	60,369	7.74
Nozzle5	33,017	4.61
Nozzle4	28,139	7.72
Nozzle1	25,705	6.61

Best RMSE = minimum across Stage 3 ablations for that fold. Once Nozzle0 is excluded, the per-variant four-fold means sit in a narrow 7.00–7.20 mm band.

Interpretation

What changed relative to the older Stage-3 ranking?

The older in-distribution automatic ranking selected `anchor_off`. Under the stricter five-fold LONO protocol with the Stage 2 teacher locked to `mu_anchor`, the best RMSE shifts slightly to `raw_uncertain_raw05_kd025`.

What is robust?

All four Stage 3 variants remain close. The largest design decision is not which Stage 3 variant wins by 0.18 mm; it is that Stage 2 should use the μ -anchor, and that Nozzle0 requires explicit discussion as an out-of-design-family fold.

How to use this result in the thesis

Report `raw_uncertain_raw05_kd025` as the LONO RMSE/MAE winner, but describe the choice as a *weak* Stage 3 preference rather than a decisive dominance claim. Keep `anchor_off` as the coverage-friendly alternative.

Cross-Architecture LONO: Same Protocol, Different Regressor

The Stage-3 LONO ablation above varies only the *regime weighting* inside the MLP family. The same five-fold LONO split is now replayed with a sparse heteroscedastic GP (SVGP) replacing the MLP, with all other axes held fixed.

Held fixed across architectures

- Five held-out families (Nozzle0/1/2/4/5); feature variant `a_plus_pressures`
- `--mode stage3 supervision (q1 synthetic + raw CDF concat)`; seed 42
- Evaluation: held-out family's uncensored CDF P50 points — point counts match MLP LONO exactly per fold

Protocol alignment. GP's built-in external eval uses the clean CDF wide table (keeps right-censored cap-hit frames). The separate evaluator `eval_gp_on_uncensored_cdf.py` reuses the MLP LONO point set so RMSEs are directly comparable. MLP row plotted: the Stage-3 LONO winner `raw_uncertain_raw05_kd025`.

SVGP vs MLP: Per-Fold (Uncensored CDF P50)

Fold (n_pts)	SVGP-Stage3			MLP raw_uncertain_raw05_kd025		
	RMSE	MAE	bias	RMSE	MAE	bias
Nozzle0 (311,149)	26.99	21.16	-18.92	34.76	27.96	-26.78
Nozzle1 (25,705)	4.88	3.72	-0.51	6.61	5.03	-4.28
Nozzle2 (60,369)	8.87	6.83	+3.50	7.74	5.73	-1.00
Nozzle4 (28,139)	5.45	4.21	-1.60	8.13	6.04	-4.91
Nozzle5 (33,017)	4.59	3.43	+1.64	5.52	4.21	+2.63
5-fold mean	10.16 $\pm$ 9.57	7.87	-3.18	12.55 $\pm$ 12.46	9.80	-6.87
4-fold (no Nozzle0)	5.95 $\pm$ 1.98	4.55	+0.76	7.00 $\pm$ 1.18	5.25	-1.89

SVGP wins point RMSE on 4 of 5 folds (all but Nozzle2). Both architectures collapse on Nozzle0 (the only first-generation injector geometry); SVGP collapses less catastrophically (-19 mm bias vs MLP's -27 mm).

Reading the SVGP–MLP Comparison

What the numbers say

- Point accuracy: SVGP wins 4 of 5 folds. 4-fold mean RMSE gap $\Delta = 1.05$ mm (5.95 vs 7.00); 5-fold gap $\Delta = 2.39$ mm carried by Nozzle0.
- Coverage: 5-fold $1\sigma/2\sigma$ GP 0.628/0.821 vs MLP 0.575/0.770; 4-fold-ex-Nozzle0 GP 0.697/0.890 vs MLP 0.696/0.923. MLP keeps only a small 4-fold 2σ edge.
- MLP's systematic negative bias (Nozzle0 -26.8 , Nozzle1 -4.3 , Nozzle4 -4.9) is consistently larger than the corresponding SVGP biases.

Why this is plausible

SVGP's Matern-5/2 ARD kernel and inducing-point variational posterior encode an implicit smoothness prior: once features move outside the training support, the posterior mean degrades toward the prior mean rather than extrapolating a learned nonlinear surface. The staged MLP, by contrast, applies `mu_anchor` on the held-out training distribution and inherits its biases into the held-out fold. The OOD ranking reversal is therefore a prior-vs-likelihood effect, not a calibration accident.

Conclusion

- ① Stage 3 LONO ablation with Stage 2 `mu_anchor` is complete: 5 folds $\times$ 4 ablations.
- ② Within the MLP family, `raw_uncertain_raw05_kd025` is the mean RMSE/MAE winner (12.55 mm / 9.80 mm); `anchor_off` gives the highest 2σ coverage (0.775). The Stage-3 spread is small (< 0.2 mm RMSE) compared with the cross-fold std.
- ③ Cross-architecture: SVGP-Stage3 wins 4 of 5 folds and the aggregate (5-fold mean 10.16 vs 12.55 mm; 4-fold-ex-Nozzle0 5.95 vs 7.00 mm). 1σ coverage is similar; MLP keeps only a small 4-fold 2σ edge.
- ④ Nozzle0 remains a design-family extrapolation failure for both architectures, but SVGP collapses less catastrophically there (RMSE 27.0 vs 34.8 mm, bias -19 vs -27 mm).
- ⑤ Thesis framing: SVGP is the stronger point predictor on both in-distribution and OOD. MLP's remaining structural value is onset-CDF supervision, the regime-aware KD chain, and ablation affordability — the staged surrogate is a regressor-agnostic instantiation of the censoring-aware learning protocol.

Artifacts:

MLP/runs_mlp/stage3_lono_ablation_20260521_092405/per_fold.csv